THE ROLLING STOCK MANUFACTURING HANDBOOK

Late-2020s engineering, production methods and quality systems

A practical technical guide to designing, building, integrating, testing and supporting premium railway vehicles.

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GB / European passenger rolling-stock baseline

Prepared as an engineering handbook. It is not a certified vehicle design, manufacturing procedure, welding procedure specification or authorisation dossier.

Document purpose and safe use

This handbook sets out a disciplined way to develop and manufacture late-2020s passenger rolling stock. It is written for a project engineering team, manufacturing engineer, production manager, quality engineer or technical director planning a new vehicle programme. The worked example is a modern intercity electric multiple unit (EMU), but the production logic can be adapted to regional, metro, locomotive and trailer-car projects.

ENGINEERING BOUNDARY A railway vehicle is a safety-critical engineered product. The sequences and control points in this handbook are practical planning instructions, not release-to-build drawings or approved numeric process parameters. Do not use generic weld settings, torque values, wheelset fits, structural allowables, brake rates, insulation clearances or dimensional tolerances from a handbook as production values. Derive them for the exact design, materials, equipment, operating envelope and applicable standards; approve them through competent engineering authority and independent assessment as required.

How to use this handbook

- Use Sections 3-6 to set the vehicle concept, requirements, interfaces and design assurance plan before ordering long-lead components.
- Use Sections 7-15 as manufacturing work packages. Convert each sequence into controlled work instructions, inspection and test plans (ITPs), risk assessments and traveller records.
- Use Sections 16-19 to plan verification, commissioning, acceptance and fleet support.
- Use the appendices as templates. Populate all bracketed or project-specific fields before authorising production.

Document assumptions

Item	Baseline used in this handbook
Market	Great Britain mainline passenger railway, with European engineering standards as a principal reference.
Vehicle	Illustrative 5-car intercity EMU, 25 kV AC overhead supply, design speed around 200 km/h.
Structure	Lightweight aluminium-alloy carbody with steel bogie frames and qualified steel wheelsets.
Traction	Distributed electric traction, modern variable-voltage/variable-frequency converters, regenerative and friction braking.
Digital	Train Control and Management System, condition monitoring, secure maintenance access and modular software.
Design life	A planning target of approximately 30-40 years, to be justified by duty cycle, maintenance and refurbishment strategy.

These are design assumptions for illustration, not a recommended procurement specification. Gauge, loading gauge, axle load, platform interface, train length, speed, braking performance, electrical compatibility and authorisation route must be confirmed for the route and operator.

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1. What “late-2020s quality” means

Late-2020s rolling stock is not defined by styling alone. It is a combination of mature lightweight structures, high-efficiency traction, digital condition monitoring, accessible interiors, carefully managed fire performance, cybersecurity, maintainable modular assemblies and data-backed quality assurance. The most convincing modern train is one whose appearance follows the engineering: clean surfaces because panels fit accurately, compact equipment because interfaces have been resolved, and quiet interiors because acoustic sources were controlled at their origin.

Quality dimension	What to design and measure
Safety and compliance	Hazard controls, structural strength, crashworthiness, fire performance, braking, accessibility, electrical protection and authorisation evidence.
Reliability	Failure Mode and Effects Analysis (FMEA), component reliability data, maintainability demonstrations, accelerated life testing where justified and service feedback.
Ride and comfort	Running dynamics, suspension tuning, noise and vibration, lighting, air quality, thermal uniformity and seat ergonomics.
Efficiency	Vehicle mass, aerodynamic drag, auxiliary energy, regenerative-brake receptivity, energy metering and maintenance energy.
Industrial repeatability	Controlled design baseline, qualified special processes, calibrated measurement, traceability and first-pass acceptance.

Quality dimension	What to design and measure
Lifecycle value	Modular repair, spares strategy, obsolescence management, refurbishment access, repair manuals and digital asset records.

CORE PRINCIPLE Design for the whole operating life, not only for the factory acceptance test. A train that is difficult to inspect, repair, clean or upgrade is not a high-quality train, even if its initial performance is impressive.

2. Reference vehicle and design baseline

2.1 Illustrative late-2020s intercity EMU

Use a reference vehicle to make the engineering workflow tangible. The following is a conceptual baseline for a standard-gauge GB intercity EMU. It is not a released design and it intentionally avoids unverified numerical values for structural loads, brake rates, electrical clearances and component fits.

Subsystem	Reference concept	Design questions to close
Formation	Five-car fixed or semi-permanently coupled unit	Train length, rescue formation, unit coupling, platform fit, passenger capacity and degraded operation.
Supply	25 kV AC overhead, 50 Hz	Pantograph compatibility, transformer arrangement, earthing, return current, route electrical limits.
Speed class	Around 200 km/h design target	Route authorisation, aerodynamic loads, running dynamics, braking distances and test-track access.
Body	Extruded aluminium alloy shell	Alloy temper, extrusion availability, joint design, crash structure, fatigue, corrosion and repair philosophy.
Running gear	Two-axle bogies, primary and secondary suspension	Axle load, wheel profile, motor suspension, curving, hunting stability, track forces and maintainability.
Traction	Distributed traction with modern converters	Power rating, duty cycle, cooling, adhesion, energy use, redundancy and electromagnetic compatibility.
Brakes	Regenerative plus electro-pneumatic friction brake	Blending, low-adhesion control, emergency brake, thermal duty, wheel-slide protection and degraded modes.
Passenger systems	Accessible modular interior, zoned HVAC and passenger information	Accessibility, fire/smoke, seat layouts, toilet systems, luggage, lighting and cleaning.
Digital	TCMS with condition monitoring and secure maintenance interface	Safety partitioning, software lifecycle, data ownership, security updates and diagnostic evidence.

2.2 Design envelope and interface register

Before detailed design, publish a Design Basis Memorandum (DBM). It should state the intended network, route constraints, operating timetable, environmental envelope, load cases, service profile, maintenance concept, customer requirements and standards baseline. Every later calculation must be traceable to a DBM assumption or an approved change.

Prepare the design basis in this order

01 Obtain infrastructure manager data: track gauge, loading gauge, platform heights and gaps, curves, gradients, axle-load limits, electrical supply characteristics, signalling interfaces, wheel/rail constraints and depot facilities.

02 Obtain operator data: annual distance, stop pattern, dwell time, passenger loading, luggage demand, cleaning regime, peak and off-peak duty, rescue policy and fleet availability target.

03 Define the vehicle operating envelope: speed, acceleration, braking, ambient temperature, humidity, rain, snow, wind, altitude, tunnel exposure, coastal salt and contamination.

04 Create a controlled interface register. Give every interface an owner, input data source, acceptance condition, verification method and due date.

05 Run a design review specifically for incompatible assumptions: platform and door interface, roof equipment and tunnel clearance, traction power and supply limits, braking and adhesion, and equipment access versus crash structure.

06 Freeze the baseline only after the engineering authority signs the DBM and the customer accepts the operational concept.

DESIGN TIP Do not begin with a styling model and attempt to fit the engineering into it. Establish the loading gauge, structural envelope, underframe depth, bogie centres, equipment volumes, door layout and service access first. Style the exterior around a proven engineering envelope.

3. Requirements, architecture and design control

3.1 Requirements management

The programme needs one controlled requirements database rather than disconnected spreadsheets and email threads. Give each requirement a unique identifier and record its source, rationale, verification method, owner, status and evidence link. Write requirements so that they can be objectively verified.

Weak requirement	Better verifiable requirement
“The train should be quiet.”	“Interior sound levels shall meet the project-defined limits at specified seats, speeds, track conditions, HVAC states and measurement positions.”
“Doors must be reliable.”	“The entrance system shall complete the defined endurance programme with failure and maintenance criteria stated in the approved door specification.”
“The body must be strong.”	“The carbody shall satisfy the defined load cases, stress/deflection limits, fatigue assessment and acceptance tests in the approved structural substantiation.”
“The train should save energy.”	“Energy use shall be measured over the agreed reference duty cycle, with auxiliary loads, regenerative energy and operating conditions declared.”

3.2 System architecture

Break the train into systems, subsystems and components. Define the mechanical, electrical, software, pneumatic, thermal and human interfaces. Create an Interface Control Document (ICD) for every boundary between suppliers or teams.

Interface	Minimum content in the ICD
Mechanical	Mounting datum, keep-out volume, mass, centre of gravity, fasteners, load cases, maintenance envelope, lifting points.
Electrical	Voltage range, current, inrush, fault protection, connectors, earthing, isolation, EMC, insulation and cable routing.
Pneumatic	Pressure envelope, air quality, flow, duty cycle, fail state, pipe material, isolation and leakage test.
Thermal	Heat rejection, cooling medium, ambient range, airflow, drainage, fire barriers and thermal interface.
Data/software	Protocol, message definition, update ownership, time synchronisation, fault coding, safety partition, cybersecurity boundary.
Passenger / human	Reach zones, clearances, access routes, usability, visual/audible indication and emergency operation.

3.3 Configuration management

Control every design release

01 Use a single Product Lifecycle Management (PLM) vault as the source of truth for drawings, models, specifications, bills of material and software baselines.

02 Assign revision and status codes: preliminary, released for manufacture, released for test, superseded or as-built.

03 Require a formal Engineering Change Request (ECR) and Engineering Change Order (ECO) for every post-freeze change.

04 Assess each change for safety, fit, function, weight, centre of gravity, fatigue, fire, EMC, software, maintainability, spares and authorisation impact.

05 Update affected drawings, inspection plans, work instructions, supplier orders, training and as-built records before the change is implemented.

06 Run a configuration audit at first article, before type testing and before delivery.

3.4 Design reviews

Review gate	Required questions	Exit evidence
Concept review	Does the train fit the operational need and infrastructure?	Approved concept, assumptions log, risk register and initial RAMS plan.
Preliminary design review	Are architecture, mass, power, structural layout and interfaces feasible?	Preliminary design, interface register, initial FMEA and verification matrix.
Critical design review	Are drawings, calculations, supplier specifications and safety controls complete?	Released design baseline, design substantiation plan and long-lead approval.

Review gate	Required questions	Exit evidence
Production readiness review	Can the factory repeat the design with controlled tooling and special processes?	Process FMEAs, work instructions, qualified procedures, ITPs and operator competence.
Test readiness review	Is the vehicle safe and instrumented for the planned test?	Approved test procedures, risk assessment, calibration records and configuration certificate.

4. Engineering calculations and digital design

4.1 CAD, CAE and digital mock-up

Use a coordinated 3D master model for vehicle layout. The model should include the carbody, bogies, equipment, cable trays, ducts, pipework, doors, seats, access panels, maintenance envelopes and lifting/removal paths. Use clash detection before drawing release. A digital mock-up is particularly valuable for roof equipment, underframe packaging and the congested interfaces around bogie pivots and cab equipment.

Digital design workflow

01 Build the vehicle skeleton model from approved datums and the loading-gauge envelope.

02 Create system zones and ownership boundaries. Publish simplified reference geometry to suppliers under revision control.

03 Integrate supplier CAD using a common coordinate system and agreed model-exchange format.

04 Run automated clash checks and manual service-access reviews. Record every unresolved clash as an action with owner and due date.

05 Run mass-property calculations at each design maturity. Track mass, axle load, centre of gravity and rotational inertia against the budget.

06 Link the 3D model to parts, drawings and assembly instructions so the digital definition and factory definition remain aligned.

4.2 Structural and fatigue substantiation

Use finite element analysis (FEA) for carbody, underframe, bogie frame, equipment brackets and critical joints. Build the model from verified geometry and material data. Document boundary conditions, mesh strategy, load cases, contacts, weld representation, assumptions and convergence checks. Do not present a colourful stress plot as proof: the calculation report must explain why the model represents the physical vehicle.

- Start with hand calculations and simplified beam or plate checks to validate the order of magnitude before trusting a large finite element model.
- Use local mesh refinement at cut-outs, brackets, weld toes and load introduction points. Demonstrate mesh convergence for critical results.
- For fatigue, use the applicable load spectrum, stress concentration treatment, joint detail class, weld quality and damage accumulation approach defined by the design standard and project method.
- Evaluate corrosion, manufacturing tolerances, residual stress and repair history where they materially affect the fatigue assessment.
- Validate important model predictions with strain gauges, displacement measurement or component tests during the physical test programme.

4.3 Vehicle dynamics and aerodynamics

Use multibody simulation to tune primary and secondary suspension, evaluate wheel/rail forces, curving, stability and ride response. Model the intended vehicle mass states, wheel profiles, track quality, speed range and relevant degraded conditions. Aerodynamic analysis should cover the nose, inter-car gaps, underbody equipment, roof equipment, cooling inlets, crosswind behaviour and tunnel pressure transients where applicable.

ENGINEERING TIP Keep the simulation configuration under the same change-control rules as the physical design. A validated simulation model is a controlled engineering asset, not an informal analysis file.

4.4 Mass and energy budgets

Budget	How to control it
Mass	Allocate mass by car and system; track design mass, growth allowance and actual weighbridge mass. Reconcile every late change.
Axle load	Calculate all load cases including passenger, luggage, consumables and equipment variation; verify with measured vehicle weights.
Energy	Model traction, braking recovery, HVAC, lighting, auxiliaries and timetable duty. Validate with metered trial data.
Thermal	Calculate heat rejection for converters, motors, transformers, batteries and HVAC condensers at worst-case ambient and duty.
Noise and vibration	Create source-path-receiver budgets and verify at component, carbody and whole-vehicle level.

5. Materials and component selection

5.1 Material selection rules

Choose materials by component function, not by fashion. Compare strength, stiffness, density, fatigue, fracture toughness, corrosion, fire behaviour, joining, repairability, supply availability, embodied carbon and recyclability. A lighter material can increase total cost if it requires complex joints, costly inspection or specialist repairs.

Component	Typical material family	Selection / control notes
Carbody extrusions	Weldable 6xxx-series aluminium alloys	Specify exact alloy, temper, extrusion quality, thickness, mechanical properties, corrosion and joining route.
Bogie frame	Qualified structural steel	Specify grade, toughness, weldability, fatigue details, heat treatment, coating and traceability.
Axles	Railway-qualified forged or rolled axle steel	Use approved axle product specification, heat-treatment route, ultrasonic examination and traceability.

Component	Typical material family	Selection / control notes
Wheels	Railway-qualified wheel steel	Specify grade, heat treatment, rim/web geometry, hardness profile, residual stress and product acceptance.
Interior panels	Fire-qualified metal sandwich or composite	Test the complete construction including core, adhesive, facing, edge finish and installed configuration.
Glazing	Railway-qualified laminated / toughened glazing as specified	Control optical quality, impact performance, retention, edge protection, bonding and replacement method.
Fasteners	Qualified high-strength or corrosion-compatible fasteners	Control grade, coating, lubrication, batch, torque/tension method and galvanic compatibility.
Seals	Railway-qualified elastomer systems	Check ozone, UV, temperature, cleaning chemical, compression set, smoke/fire and replacement interval.

5.2 Incoming material inspection

Incoming goods procedure

01 Confirm purchase order and approved supplier match the current bill of material and drawing revision.

02 Check delivery condition, identification marks, packaging, damage, contamination and transit protection.

03 Verify material certificates and batch/heat numbers against the specification. Check that the certificate type is the one required by the purchase order.

04 Perform required dimensional, visual and material checks using calibrated equipment and approved sampling plans.

05 Quarantine unverified or non-conforming goods. Do not allow a missing certificate to be solved by informal substitution.

06 Apply durable part and batch identification before material is cut or split into smaller items. Transfer traceability to offcuts when they remain usable.

07 Release material in the ERP/MES only after the authorised quality function records acceptance.

5.3 Dissimilar material and corrosion control

Where aluminium, stainless steel, carbon steel, copper and composites meet, assess galvanic corrosion, moisture traps, crevice corrosion, thermal expansion and repairability. Use suitable isolation barriers, sealants, coatings, drainage and compatible fasteners. Detail every hidden cavity so water cannot collect unnoticed. Provide inspection access to likely corrosion hot spots.

- Avoid bare dissimilar-metal contact in wet zones unless the joint has been specifically designed and qualified for that environment.
- Do not seal a cavity without a drainage, venting and inspection strategy.
- Specify coating repair systems and surface-preparation instructions for depots, not just the original factory paint system.
- Record the exact material and coating system used at interfaces that will later be concealed.

6. Factory planning and manufacturing system

6.1 Factory layout

Design the factory around a one-way flow of materials and vehicles. Keep dirty cutting and surface-preparation areas away from clean electrical integration and interior finishing. Create a clear quarantine route for non-conforming parts. Plan heavy lifting, fire separation, extraction, hazardous-substance control, compressed air, electrical supplies and emergency egress before installing machines.

Area	Core equipment / capability	Key facility controls
Goods-in and material store	Racking, profile handling, plate storage, barcode/RFID tracking	Material segregation, humidity control where needed, lifting plans, certificate verification.
Cutting and machining	CNC saws, milling, drilling, laser/plasma where suitable, deburring	Fume extraction, guarding, chip control, calibration and tool management.
Carbody fabrication	Long-bed fixtures, robotic/manual welding cells, FSW where qualified	Weld fume capture, fire prevention, fixture verification and process qualification.
Bogie and wheelset shop	Machining, press equipment, balancing, alignment, inspection	Heavy-load controls, clean bearing assembly, guarded presses and NDT access.
Paint and surface treatment	Preparation bays, controlled spray booths, curing and inspection	Ventilation, hazardous-area classification, fire protection and environmental controls.
Electrical integration	Cable cutting, crimping, harness boards, test benches	ESD controls where required, segregation, electrical safety and version control.
Interior fit-out	Panel, seat, floor, glazing and door installation stations	Cleanliness, material protection, ergonomic lifting and fire-compliant materials.
Final assembly and test	Pit access, lifting jacks, diagnostic rigs, static test points	Safe access, isolation, test instrumentation and configuration control.
Metrology and NDT	CMM, laser tracker, UT, MT, PT, radiography as applicable	Controlled environments, qualified inspectors and equipment calibration.

6.2 Production system

Use Manufacturing Execution System (MES) work orders connected to the PLM bill of material. Each station should receive an approved work pack, scan the vehicle/assembly identity, confirm parts and revision, record the operator and equipment, complete the operation and attach inspection evidence. A step should not close while mandatory hold points remain open.

6.3 First article and rate production

Move from prototype to repeat production

01 Build a first article using the intended production tools, fixtures, suppliers and work instructions. Do not treat a hand-built demonstrator as evidence that the rate process is capable.

02 Inspect the first article against the full drawing and process plan, not only a small sample of dimensions.

03 Run process capability studies for critical-to-quality characteristics where the data and process are suitable. Establish control limits from evidence, not arbitrary targets.

04 Complete a production readiness review covering tooling, operator qualification, maintenance, spare tooling, calibration, supplier capacity and inspection capacity.

05 Authorise rate production only when non-conformities are closed or formally risk-assessed and the design baseline is stable.

06 Continue first-off inspection after significant tooling, supplier, material, software or process changes.

FACTORY TIP Make it physically difficult to assemble the wrong revision. Use keyed connectors, part scanning, fixture poka-yoke, visual work instructions and software interlocks. Do not rely on memory or a final inspector to catch avoidable assembly errors.

7. Carbody manufacture: from stock to accepted shell

The carbody is the primary structural shell. Its manufacturing process needs stable reference datums, repeatable fixtures, controlled joining and independent dimensional inspection. The project should issue a dedicated Carbody Manufacturing Plan (CMP) that links each operation to the structural drawing, weld map, inspection requirement and release authority.

7.1 Preparation and datum strategy

Before cutting material

01 Confirm the released-for-manufacture carbody drawing set, material list, weld map and assembly sequence. Verify no unresolved engineering change affects the shell.

02 Identify the master longitudinal, transverse and vertical datums. Mark datum transfer points on fixtures and measurement targets.

03 Inspect the assembly jig for damage, wear, looseness, contamination and calibration status. Record its identification and verification status.

04 Check material heat/batch identity and orientation. Mark each blank with a traceable ID that survives cutting and forming.

05 Confirm the approved cutting program, tool condition, nesting file and revision. Make a first-off cut and check its geometry before releasing the batch.

06 Deburr edges and remove contamination using the approved method. Protect finished surfaces from scratches and embedded ferrous particles.

7.2 Underframe assembly

Underframe manufacturing sequence

01 Load longitudinal members, cross-members, end structures and equipment support members into the master fixture in the specified order.

02 Locate each component on the drawing-defined datums. Use calibrated gauges or digital measurement to verify position before tack operations.

03 Fit temporary restraints and clamps using the fixture plan. Do not force misaligned parts into position; investigate root cause and correct the part or fixture through an approved disposition.

04 Complete tack joining using the qualified procedure and sequence. Recheck geometry after tack-up because local restraint can introduce distortion.

05 Execute permanent joining in the approved balanced sequence. Follow the qualified procedure for joint preparation, consumables, heat input, interpass controls and environmental conditions.

06 Allow controlled cooling and release restraints in the specified order. Do not accelerate cooling unless the qualified process explicitly permits it.

07 Perform visual and specified NDT inspections. Record weld IDs, inspector, equipment, technique and disposition.

08 Measure the underframe datums, diagonals, twist, mounting faces and critical interfaces. Compare against the released drawing and measurement plan.

09 Complete approved local machining only after the joining and stress/distortion strategy allows it. Reinspect machined faces and holes.

10 Issue a signed underframe release record and protect all datum surfaces for subsequent shell assembly.

7.3 Sidewalls, roof and end structures

Manufacture wall and roof subassemblies in dedicated fixtures. For extruded panels, control profile identity, extrusion straightness, joint gap, panel orientation and local surface condition. Door and window apertures require local reinforcement designed around fatigue and crash load paths. Avoid late field-cutting of structural openings.

Sidewall and roof work sequence

01 Verify the extrusion/panel schedule and identify each part against the assembly drawing.

02 Place components in a checked fixture and confirm gap, flushness, sweep and key aperture dimensions before joining.

03 Install reinforcement and doubler features in the specified sequence; confirm that the detail matches the stress and fatigue substantiation.

04 Join using the qualified process. Monitor fixture temperature, restraint, tool condition and sequence as required by the process plan.

05 Inspect the completed panel for distortion, surface damage, joint condition and aperture geometry.

06 Protect finished panels from handling marks and moisture. Store on engineered supports that do not introduce permanent set.

7.4 Shell erection and closure

Build the full carbody

01 Position the underframe in the main shell fixture and verify master datums.

02 Install sidewall subassemblies in the planned order. Use temporary locating devices and check symmetry before permanent joining.

03 Install end structures and roof assembly. Maintain access for internal joining and inspection before closing inaccessible cavities.

04 Complete the structural joining sequence while recording each weld or joining operation against the weld map.

05 Perform an intermediate survey before fitting secondary brackets and equipment supports. Correct any deviation under engineering disposition.

06 Install structural closures only after required internal inspections, fire-barrier checks and concealed-joint records are complete.

07 Carry out the full carbody dimensional survey using a laser tracker or equivalent calibrated system.

08 Compare survey results to the approved datum report. Rework only under an approved instruction and remeasure the affected interfaces.

09 Complete structural NDT, visual inspection, drainage checks and surface preparation release.

10 Issue the carbody acceptance certificate before paint or fit-out begins.

7.5 Carbody dimensional control

Characteristic	Control method	What to record
Overall geometry	Laser tracker / photogrammetry / calibrated gauges	Datum setup, instrument calibration, point cloud or measured points, deviations.
Bogie interface points	Dedicated reference tooling and survey	Centre locations, mounting plane, perpendicularity and relationship to vehicle datum.
Door/window apertures	Template / CMM / laser measurement	Opening dimensions, diagonals, flatness and local reinforcement position.
Roof and underframe mounts	CMM / laser tracker / gauges	Hole pattern, mating plane, angular orientation and accessibility.
Twist and straightness	Defined datum survey method	Measurement conditions, support condition, temperature and calculations.

DIMENSIONAL TIP Do not specify one generic tolerance for “the carbody”. Establish tolerances by function: safety/load path, bogie alignment, door sealing, glazing fit, equipment interface, exterior styling and cosmetic panel match. Use a formal tolerance stack-up for each critical interface.

8. Welding, joining and structural inspection

8.1 Special-process control

Welding and other special processes cannot be fully verified by looking at the finished part. Control the process itself. Establish a welding coordination function, approved welding manufacturer capability, qualified procedures, qualified personnel, consumable control, equipment checks, traceable weld maps and inspection requirements. Apply the current railway welding standard and the customer’s weld-quality requirements.

8.2 Welding procedure release

Before production welding

01 Classify each welded component and joint according to its safety significance and the applicable EN 15085 requirements.

02 Prepare joint details showing material, thickness, joint type, access, position, preparation, backing arrangement and inspection class.

03 Select a joining process suited to the material and geometry. Consider distortion, fatigue, corrosion, access and repairability.

04 Qualify the welding procedure through the required procedure qualification route. Record essential variables and the qualified range.

05 Confirm that each welder/operator holds valid qualification for the process, material group, position and thickness range involved.

06 Issue a controlled WPS at the point of use. Confirm equipment identification, calibration and consumable batch.

07 Run a first-off production joint and inspect it before authorising the same setup for the rest of the batch.

08 Record production welds against unique weld IDs and retain inspection and repair history.

8.3 Arc welding: practical controls

For aluminium or steel arc welding, the WPS must define the process-specific variables. The work instruction should make the operator check them at setup and at the specified frequency. Avoid copying settings from another thickness, alloy, joint or machine.

Control	Operator / supervisor instruction
Joint preparation	Check bevel, root face, fit-up, cleanliness, oxide removal where relevant and approved gap range before striking an arc.
Consumables	Verify exact classification, diameter, batch, storage condition, shelf life and issue/return procedure.
Equipment	Verify machine ID, maintenance status, wire feed, gas supply, torch condition, earth return and calibration/verification status.
Process parameters	Set current, voltage, travel speed, wire feed, gas flow, preheat and interpass controls only from the released WPS.
Environment	Apply the WPS limits for wind, draught, temperature, humidity, condensation and surface cleanliness.
Sequence	Follow the approved weld sequence and restraint plan. Pause at defined checks to measure distortion.
After welding	Mark weld ID, complete visual inspection and specified NDT; record repairs and re-inspect repaired areas.

8.4 Friction stir welding (FSW)

FSW is useful for selected aluminium extrusions and long straight joints where the material, geometry and production rate justify the specialist tooling. The tool generates frictional heat and plasticises material without conventional fusion melting. A production FSW procedure must be qualified for the exact alloy/temper, thickness, joint design, machine, tool, backing support and parameter window.

FSW production preparation

01 Confirm the joint is suitable for FSW and has been substantiated for static strength, fatigue and defect tolerance.

02 Check tool identity, tool condition, shoulder/pin geometry, wear limit and traceable service history.

03 Verify clamping, backing support, joint gap, alignment and machine coordinate calibration.

04 Load only the approved parameter recipe for the exact joint. The recipe must define the controlled variables and alarms; do not improvise rotation, traverse, plunge or tilt values.

05 Run a representative first-off coupon/panel and apply the approved inspection and macro-section or destructive qualification regime.

06 During production, monitor the parameters and alarms that the qualified procedure identifies as essential. Stop and quarantine the panel if a controlled variable leaves the approved window.

07 Inspect start/stop regions, run-on/run-off tabs and the full joint using the qualified inspection method. Record machine trace and weld map.

08 Allow only authorised repairs. FSW repair design and procedure must be separately qualified.

8.5 NDT and non-conformance

Select visual testing, penetrant testing, magnetic particle testing, ultrasonic testing or radiography according to material, joint type, access, defect orientation and risk. Define the inspection technique, extent, acceptance criteria, personnel qualification and report template before production starts. NDT is not a substitute for process control.

HOLD POINT If a safety-critical weld fails acceptance, stop the affected operation, identify the full population potentially made under the same condition, quarantine it and open a non-conformance report. Do not repair, blend or accept by concession without authorised engineering disposition and documented reinspection.

9. Bogies, wheelsets and running gear

9.1 Bogie design and procurement

Bogie design is a coupled structural and dynamics problem. The frame, wheelsets, bearings, suspension, traction linkages, gearbox, brake equipment and carbody interface must be assessed as a system. Define the vehicle mass and loading cases, track geometry, operating speed, wheel profile, axle load, maintenance interval and degraded operating requirements before freezing the bogie.

9.2 Bogie frame manufacturing

Bogie frame route

01 Review the released bogie frame drawing, structural substantiation, material specification, weld map and inspection plan.

02 Inspect raw plate and sections for material identity, thickness, condition, straightness and certificate conformity.

03 Cut and machine parts using controlled CNC programs; deburr and mark each part with traceable identification.

04 Load parts into the verified bogie fixture. Check datum alignment, diagonal dimensions and mounting-face geometry before tack joining.

05 Join using qualified welding procedures and the approved balanced sequence. Monitor heat input and distortion as defined by the WPS and process plan.

06 Perform the specified post-weld treatment only if it is included in the qualified design and procedure. Do not introduce unapproved heat treatment or straightening.

07 Complete visual and specified NDT inspections. Record weld IDs and any repair.

08 Machine critical interfaces only at the planned stage and maintain datum continuity between machining and final survey.

09 Perform final dimensional inspection and structural acceptance. Apply corrosion protection after all required inspections and machining are complete.

10 Release the frame with a dossier linking material heats, welders, procedures, NDT and final survey.

9.3 Wheelset manufacture and assembly

Wheelsets and axles are safety-critical. They should normally be purchased from a specialist, approved manufacturer unless the rolling-stock builder has the complete forging/rolling, heat treatment, machining, testing and certification capability. The vehicle manufacturer remains responsible for specifying, verifying and tracing the supplied product.

Wheelset receiving and assembly controls

01 Verify axle and wheel certificates, heat/batch identity, product grade, heat treatment, dimensional reports and NDT records against the approved specification.

02 Inspect delivery condition and preserve identification marks. Quarantine any component with damaged surfaces, missing certificates or uncertain history.

03 Check machined wheel seats, axle journals, fillets, shoulders, surface finish and dimensions using calibrated equipment.

04 Confirm the approved assembly method and press-fit specification for the exact wheel/axle design. Confirm press calibration and data acquisition before assembly.

05 Perform assembly under controlled cleanliness and temperature conditions. Record the complete press-fit curve or equivalent process record where required.

06 Check wheelset geometry, wheel back-to-back, runout, concentricity, wheel profile and axle dimensions to the approved drawing and standard.

07 Carry out specified post-assembly NDT and balancing/verification steps.

08 Apply permanent wheelset identity and link it to the axle and wheel serials, component records and maintenance history.

09 Release the wheelset only after the responsible quality authority signs the complete dossier.

DO NOT GUESS Wheel press-fit interference, press force/curve limits, wheel back-to-back dimensions, axle journal tolerances and wheel-profile limits are design- and standard-specific. Use the controlled component specification and current applicable standard; never copy a value from a different gauge, axle or wheel design.

9.4 Suspension and bogie integration

- Install primary and secondary suspension parts using the correct part numbers, orientation and batch/serial traceability.
- Verify spring rates, damper characteristics, air-spring specification and any matched-pair requirements using supplier certificates and incoming tests.
- Use the specified fastener strategy, approved torque/tension method and calibrated tools. Apply witness marks only after the joint has passed the approved tightening verification.
- Check brake rigging, cable routing, traction motor/gearbox alignment, pipe supports, sensor installation and clearances over the full movement envelope.
- Perform a bogie geometry survey and functional checks before fitment to the vehicle.

9.5 Bogie-to-carbody installation

Installation sequence

01 Confirm carbody and bogie interfaces have been released and all transport restraints are removed.

02 Inspect lifting points, lifting beam certification, jack capacity and exclusion zone. Use an approved lift plan.

03 Identify bogie orientation and match it to the vehicle installation drawing. Confirm wheelset and bogie serial numbers.

04 Raise or lower the carbody/bogie using the approved sequence. Keep personnel clear of crush zones and never work under an unsupported suspended load.

05 Install pivot, centre bowl, traction links, yaw dampers, air springs, brake connections, earthing bonds and electrical connectors as specified.

06 Complete fastener verification using controlled procedures, then inspect clearances, hose routing, harness support and articulation.

07 Complete air leak, brake continuity, sensor and electrical checks before the vehicle leaves the installation bay.

10. Traction, high-voltage and low-voltage systems

10.1 Electrical architecture

The electrical design should separate traction power, auxiliary power, control power and safety-critical supplies. Define normal, degraded, emergency and isolated states. Create a power single-line diagram, earthing/bonding diagram, cable schedule, protection coordination study, EMC plan and electrical safety case.

System	Engineering controls
Current collection	Pantograph interface, roof clearances, high-voltage isolation, earthing, surge protection and contact-force compatibility.
Transformer / input	Thermal duty, insulation coordination, oil or dry-type safety strategy, cooling, protection and maintenance access.
Traction converter	Semiconductor choice, switching strategy, cooling, DC link, fault handling, EMC, maintainability and obsolescence.
Traction motor	Power/speed/torque duty, thermal class, insulation, bearings, suspension, cooling and monitoring.
Auxiliary converter	HVAC, compressors, lighting, doors, battery charging and emergency load priorities.
Battery	Chemistry, energy/power, thermal management, containment, BMS, fire response, isolation and safe maintenance.
Low-voltage distribution	Protection coordination, cable sizing, voltage drop, segregation, labels and maintainable fuse/contactor access.
Earthing and bonding	Traction return, protective bonding, equipotential bonding, stray-current strategy and continuity test points.

10.2 High-voltage assembly work

ELECTRICAL SAFETY High-voltage railway work must be performed only by authorised, competent persons under the operator/manufacturer electrical safety rules. This handbook is not an isolation procedure. Use the approved safe system of work, permit, proving unit, test instrument, earthing and discharge arrangements for the exact equipment.

Engineering and assembly controls

01 Approve the electrical single-line diagram, isolation philosophy, protection study and cable schedule before installation.

02 Check all high-voltage equipment for transport damage, contamination, moisture ingress, certification and correct part identity.

03 Install equipment on approved mounts and torque/tension joints using the released joint specification.

04 Route high-voltage cables separately from sensitive low-voltage, data and sensor cables as required by EMC and insulation design.

05 Maintain bend radius, support spacing, creepage/clearance, fire barriers, grommets, edge protection and segregation from hot or moving parts.

06 Fit labels, warning signs, interlocks and covers before testing. Confirm access panels cannot expose live parts during normal operation.

07 Complete continuity, insulation, bonding, protection and functional tests using approved procedures and calibrated instruments.

08 Record serial numbers, cable IDs, test results, test instrument IDs and the final as-built routing.

10.3 Cable harness manufacture

Harness shop work instruction

01 Generate harness data directly from the controlled cable schedule and wiring diagram; confirm drawing revision at the workbench.

02 Cut wire to controlled length with the specified allowance and maintain wire ID at both ends.

03 Strip insulation using the specified tooling and setting. Reject strands that are cut, nicked or damaged.

04 Crimp using the qualified terminal, die, applicator and wire combination. Perform first-off pull/section tests at the approved frequency.

05 Fit connectors, backshells, seals and strain relief in the correct orientation. Observe pin-out and keying carefully.

06 Build the harness on a full-size board where practical; verify branch lengths, clips, bend radii, labels and connector orientation.

07 Perform continuity, insulation and pin-to-pin checks. Use automated test equipment where appropriate.

08 Protect the harness for storage and transport. Do not allow it to be dragged across sharp edges or crushed under stored parts.

10.4 Modern traction technologies

Late-2020s traction packages may use silicon-carbide power devices, compact liquid-cooled converters, high-efficiency motors and more sophisticated energy-management software. These technologies can reduce losses and packaging volume, but bring thermal, EMC, insulation, repair and spares implications. Compare whole-system efficiency and lifecycle cost rather than selecting a semiconductor by headline switching efficiency alone.

11. Braking, pneumatics and safety systems

11.1 Brake concept and design verification

The brake design must establish stopping and immobilisation performance across the defined speed, load, gradient, adhesion, supply and fault envelope. A typical modern EMU blends regenerative braking with friction braking; it must transition predictably when the electrical network cannot accept regenerated energy or when a traction component is unavailable.

- Define normal service, full service, emergency, parking, holding and rescue brake modes.
- Define the control hierarchy and ensure an independent safety path for emergency braking as required by the vehicle architecture.
- Model brake energy, power, thermal capacity, wear and repeated-stop duty. Include worst-case operating sequences and degraded states.
- Specify wheel-slide protection and low-adhesion response, and verify that it does not create unacceptable stopping-distance or wheel-damage outcomes.
- Define brake status feedback, fault detection, isolation rules and driver indications.

11.2 Brake assembly controls

Install and verify the brake system

01 Inspect brake discs, pads, calipers, actuators, valves, sensors and pipework against approved part numbers and certificates.

02 Check mounting faces, fastener condition, clearances and orientation before installation.

03 Install disc/caliper assemblies using the controlled torque/tension and locking method specified by the design.

04 Install pneumatic pipework with approved material, supports, bend radius, protection and cleanliness controls.

05 Keep open pipe ends capped. Prevent swarf, sealant fragments, water and cleaning fluid from entering the pneumatic system.

06 Perform leak and pressure tests using the approved test pressure, duration and acceptance criteria for the specific circuit.

07 Complete brake continuity, command response, sensor calibration, wheel-slide protection and failure-mode checks.

08 Record each test against car, bogie, brake unit and software configuration.

11.3 Pneumatic cleanliness

Air quality has a direct effect on valves, actuators and brake reliability. Define the required air cleanliness, water and oil limits from the equipment specification. Use clean capped hoses, verified dryers and filters, controlled pipe preparation and flushing/purging procedures. Never use unregulated workshop air to blow debris into a completed circuit.

11.4 Safety systems and emergency features

The project safety plan should include hazard analysis for emergency braking, door interlocks, emergency lighting, passenger alarm, evacuation, fire detection where specified, cab vigilance, train protection interfaces and rescue coupling. Demonstrate that each safety function reaches its defined safe state under loss of power, broken wire, sensor fault, communication fault or controller reset as appropriate.

12. Doors, glazing, HVAC and passenger interior

12.1 Passenger doors

Passenger doors are high-cycle safety-critical equipment. Buy or design them as a complete qualified system, including leaves, guides, actuators, controls, obstacle detection, seals, emergency release, indication and mechanical locking. Ensure that door suppliers receive the carbody interface drawing, platform envelope and required duty cycle early.

Door installation and acceptance sequence

01 Inspect the door frame opening and record its geometry before installing the door cassette.

02 Confirm the cassette, leaves, seals, drive and control hardware match the approved configuration.

03 Install the cassette on the defined datums; shim only as allowed by the approved installation instruction.

04 Adjust the door leaf position, sealing compression, parallelism and running clearances using the manufacturer procedure.

05 Install wiring and sensors with protected routes, correct connector keying and strain relief.

06 Run the defined operational and obstacle-detection tests, including local and train-wide commands.

07 Run emergency release and isolation tests, then restore the system and verify normal operation.

08 Complete an endurance test programme and record cycle count, faults, adjustments and replaced components.

12.2 Windows and glazing

Treat glazing as a structural and fire-related installation, not a cosmetic component. Define glass build-up, optical requirements, retention, bonding or gasket system, edge protection, impact performance, thermal loads and replacement method. Use the approved surface preparation, adhesive batch control and cure conditions. Record environmental conditions and adhesive lot for every bonded installation.

12.3 HVAC and ventilation

HVAC installation and commissioning

01 Confirm the airflow, cooling and heating design loads for each car and zone, including passenger load, solar gain, door opening and external ambient conditions.

02 Inspect HVAC modules, filters, fans, coils, dampers, sensors, drains and fire/smoke interfaces on receipt.

03 Install equipment on designed isolators and mountings. Verify maintenance removal paths before final closure of surrounding panels.

04 Connect ducts and drains with smooth transitions, sealed joints, appropriate insulation and access to filters and coils.

05 Pressure/leak test the relevant ducts or refrigerant circuits under the approved procedure and applicable refrigerant safety rules.

06 Balance airflow by measurement at specified outlets; record air volume, temperature, control mode and ambient conditions.

07 Test heating/cooling, fresh-air control, recirculation, filtration, condensate drainage, fault response and passenger noise.

08 Check thermal uniformity in representative load conditions and update control tuning only through software change control.

12.4 Interior fit-out

Use replaceable modules and standardised attachment interfaces. Prioritise robust seat structures, replaceable upholstery, cleanable surfaces, accessible fasteners, anti-vandal detailing and consistent panel gaps. Avoid hidden cable joints behind bonded panels where a removable service hatch would be more maintainable.

Interior element	Manufacturing and quality instructions
Floor	Verify substrate flatness and dryness; use approved adhesive, seam and edge details; inspect slip resistance and cleanability.
Seats	Verify frame identity, structural test evidence, anchorage, cushion fire performance and replaceable trim. Check pitch and clearances.
Panels	Verify fire certificates for the complete assembly, not just a raw resin or face sheet; inspect clip engagement and service removal.
Lighting	Use modular LED luminaires, replaceable drivers, consistent colour temperature and accessible connectors; test dimming and emergency modes.
Passenger information	Check display sightlines, accessibility, brightness, audio intelligibility and message consistency with train control.
Toilets	Check tank sealing, drainage, ventilation, flush cycle, leak detection, service isolation and access for cleaning and repair.

Interior element	Manufacturing and quality instructions
Luggage and tables	Verify retention, passenger impact risk, sharp edges, load capacity and attachment strength.

12.5 Accessibility and passenger ergonomics

Run accessibility reviews with representative users and the operator, not only a drawing review. Evaluate boarding gap and height, wheelchair circulation, accessible toilet and spaces, handrail reach, visual contrast, announcement audibility, signage, luggage movement and emergency evacuation. The accessibility solution must be coherent with the platform interface and operational assistance model.

13. Surface protection, exterior finish and final fit

13.1 Surface preparation and coating

Set the coating system by substrate, exposure class, cleaning chemicals, expected graffiti removal, UV, salt and repair conditions. The coating supplier should provide a complete approved system with preparation grade, pretreatment, primer, build coat, topcoat, curing envelope and repair method.

Paint-shop work instruction

01 Confirm that all structural inspections and concealed-joint checks are complete before surface treatment.

02 Protect machined interfaces, electrical earth points, identification plates, seals, glazing seats and other no-coat areas.

03 Prepare the surface using the substrate-specific approved method. Check cleanliness, profile and contamination before coating.

04 Record booth temperature, relative humidity, surface temperature and dew point margin as required by the coating supplier specification.

05 Apply each coat at the specified film build and cure conditions. Record product batch, mix ratio, pot life and operator.

06 Measure dry-film thickness at defined locations and inspect adhesion, coverage, runs, pinholes and colour.

07 Repair defects only using the approved repair procedure. Reinspect the repaired area and record it.

08 Protect the finish during onward transport and interior fit-out. Use non-marking supports and clean protective film compatible with the coating.

13.2 Exterior late-2020s design language

- Use a clean, coherent front-end geometry with a clear visual identity, but preserve crash structure, driver sightlines, coupling access, demisting, lighting and aerodynamic requirements.
- Prefer fewer, accurately fitted exterior panels over many decorative pieces. Tight, even panel interfaces communicate build quality and reduce dirt traps.
- Integrate roof equipment into low-profile fairings only where cooling, maintenance, fire access and electrical clearances remain satisfactory.
- Use durable, repairable colour and graphic schemes. Design livery transitions around replaceable panel boundaries so repairs do not require large-area repainting.
- Specify corrosion-resistant fasteners and concealed drainage; do not allow styling caps to hide inspection points or water traps.

13.3 Final fit and finish inspection

Inspection zone	Acceptance focus
Exterior panels	Flushness, gap consistency, colour match, surface waviness, fastener security, sealant finish and drainage.
Glazing	Optical condition, edge finish, bond line, gasket seating, no water ingress and correct markings.
Doors	Leaf alignment, seals, flushness, emergency labels, opening/closing sound and correct status indication.
Roof	Access covers, fairings, cable glands, lifting provisions, water paths and HV warning labels.
Underframe	Pipe/cable support, loose items, coating coverage, drainage, guards and identification.
Interior	Panel gaps, upholstery, trim, cleanliness, smell, sharp edges, loose parts, lighting and accessible service hatches.

14. Software, TCMS, diagnostics and cybersecurity

14.1 TCMS architecture

The TCMS coordinates traction, braking, doors, HVAC, auxiliary power, passenger information and diagnostic functions. It must be designed with clearly defined safety partitions, communication boundaries, degraded modes, time synchronisation, event logging and software configuration management.

- Maintain a system architecture diagram showing controllers, networks, gateways, I/O, power sources and safety functions.
- Define data dictionaries and message interfaces with supplier ownership and version control.
- Ensure that loss of a non-safety service network does not defeat a safety function.
- Create fault codes that identify the system, component, severity, operator action and diagnostic evidence needed.
- Ensure maintenance updates are authenticated, authorised, logged and reversible to a known-good configuration where safe and approved.

14.2 Software lifecycle

Software release workflow

01 Allocate software requirements from the system-level specification and hazard analysis.

02 Classify the software and functions according to the safety lifecycle and applicable standards.

03 Create a software development plan covering architecture, coding rules, reviews, verification, validation, tools and configuration management.

04 Use controlled source repositories, peer review, automated tests, static analysis and traceability from requirement to test.

05 Perform integration testing with representative hardware and communication networks; include fault injection for defined failure cases.

06 Freeze a tested software baseline for vehicle test. Record version, checksum, hardware compatibility and configuration.

07 Control every subsequent update through impact assessment, regression testing, approval and fleet rollout management.

14.3 Condition monitoring

Monitor only what can be interpreted and acted upon. Useful channels may include bearing temperature, vibration, motor temperature, converter status, brake wear, compressor duty, door cycle count, HVAC performance and battery state. Set thresholds from supplier limits, baseline measurements and service data. A diagnostic should identify a likely failure mode and recommended inspection, not simply generate an alarm.

14.4 Cybersecurity

- Inventory every onboard asset, software component, network link and remote maintenance path.
- Separate operational technology networks from passenger Wi-Fi and non-essential entertainment networks.
- Use role-based access, authenticated maintenance sessions, secure boot or equivalent platform controls where applicable, and auditable update mechanisms.
- Define vulnerability handling, security patch evaluation, supplier notification and end-of-support arrangements.
- Test recovery from corrupted updates, loss of communications and compromised maintenance credentials within an approved cyber test environment.

LATE-2020S TIP Build the diagnostic data model early. Retrofitting sensors and data fields after the vehicle is built is expensive, and a large volume of poorly structured telemetry is not the same as useful predictive maintenance.

15. Production quality, traceability and supplier control

15.1 Quality plan

Create a Project Quality Plan aligned to ISO 22163 and the customer contract. It should identify the quality organisation, document control, inspection independence, supplier controls, special processes, calibration, non-conformance, corrective action, audit programme and final record pack.

15.2 Inspection and Test Plans

Each work package should have an ITP with a defined sequence, characteristic, method, acceptance reference, frequency, responsible party and hold/witness point. A hold point prevents the work from proceeding until the designated authority releases it. A witness point gives the nominated party an opportunity to observe the operation under the agreed notification rules.

ITP field	What the planner must write
Operation ID	Unique work-step identifier and location in process route.
Characteristic	Exactly what is being checked: datum location, weld, insulation, leak, function, surface, software version, etc.
Method	Drawing inspection, calibrated gauge, NDT method, electrical test, functional test or visual criteria.
Acceptance reference	Drawing revision, standard clause, approved specification, test procedure or customer requirement.
Extent / frequency	100%, sample plan, first-off, lot-based or periodic; define the lot and sampling logic.

ITP field	What the planner must write
Responsibility	Operator self-check, production inspector, quality inspector, supplier inspector, independent assessor.
Point type	Surveillance, review, witness or hold. State release authority and notification lead time.
Record	Report form, equipment ID, calibration, result, signature, date and non-conformance reference.

15.3 Non-conformance and corrective action

Non-conformance workflow

01 Identify and contain the non-conforming item or process. Physically and digitally prevent unintended use.

02 Raise a Non-Conformance Report (NCR) with part ID, serial/batch, drawing revision, process step, evidence and quantity potentially affected.

03 Investigate the root cause. Separate the immediate symptom from the process/system cause.

04 Obtain authorised disposition: rework, repair, use-as-is concession where legally and technically permissible, return to supplier or scrap.

05 Issue controlled rework instructions and inspect the result against defined acceptance criteria.

06 Assess the same condition across other vehicles, assemblies and supplier lots.

07 Implement corrective action, verify effectiveness and update FMEA, work instructions, training or supplier controls as necessary.

15.4 Supplier quality

Classify suppliers by product safety significance, technical complexity, process specialisation and supply risk. Conduct capability assessments before award and surveillance audits during production. Critical suppliers should provide a quality plan, approved process route, source inspection access, change notification, full traceability, test certificates and a defined non-conformance response time.

15.5 Calibration and measurement system

- Maintain a master register of all measuring equipment, calibration status, range, resolution, location and owner.
- Use measurement uncertainty appropriate to the tolerance being verified. A gauge that cannot resolve the required limit is not suitable simply because it has a current calibration sticker.
- Verify fixtures and master gauges at defined intervals and after damage, repair or relocation.
- Record the instrument ID with each critical inspection result.
- When a device is found out of tolerance, assess whether previous product inspections are invalid and identify the affected product population.

16. Static verification, integration and type testing

16.1 Test strategy

Build the Verification and Validation Plan before design freeze. For each requirement, specify whether it will be verified by analysis, inspection, demonstration or test. Allocate testing at component, subsystem, vehicle and train level. Use a requirement-to-evidence matrix so that no requirement is left without an acceptance route.

Test level	Typical activities	Release condition
Component	Material tests, supplier tests, converter bench, motor test, door endurance, brake component tests	Component certificate and approved test report.
Subassembly	Bogie, wheelset, HVAC module, electrical rack, door cassette, harness	Subassembly inspection and functional release.
Vehicle static	Dimensions, mass, insulation, bonding, brake continuity, doors, HVAC, lights, TCMS, alarms	Static test certificate; open safety-critical defects closed.
Train integration	Coupled train communications, traction/brake coordination, degraded modes, passenger systems	Integrated test report and configuration record.
Dynamic	Running behaviour, ride, noise, braking, energy, current collection and route compatibility	Approved test report against defined acceptance criteria.
Operational trial	Service-like duty, reliability growth, maintainability and passenger operations	Trial close-out and authorisation/acceptance evidence.

16.2 Static vehicle test sequence

Recommended static test order

01 Confirm the vehicle is mechanically complete, correctly supported and in the approved test configuration.

02 Complete visual and dimensional walkdown, mass/weight verification and equipment identification audit.

03 Complete protective bonding, earthing, continuity, insulation and electrical protection tests under the approved electrical safety procedure.

04 Power the low-voltage control system and verify controller configuration, network health, time synchronisation and diagnostic logs.

05 Test cab controls, safety interlocks, emergency devices, communications and driver indications.

06 Test each door locally and from the train-level command, including obstruction detection, isolation and emergency release.

07 Test lighting, passenger information, audio, HVAC, toilets, accessibility features and auxiliary services.

08 Test brake continuity, pressure build-up, leak rate, applied/released state indication, emergency brake command and parking/holding functions.

09 Run integrated fault simulations for the approved list of sensor, network, power and component faults.

10 Close defects, repeat affected tests and issue the static vehicle test report with the final software and hardware configuration.

16.3 Test instrumentation

Use sensors and data acquisition sized for the expected signal range, frequency and uncertainty. Typical instrumentation may include accelerometers, strain gauges, temperature sensors, pressure transducers, current/voltage probes, wheel force measurement, GPS/time synchronisation and calibrated sound level meters. Create an instrumentation plan that records sensor ID, location, orientation, calibration, sample rate, filtering and channel name.

16.4 Test safety

- Approve a task-specific risk assessment and test plan before energising or moving the vehicle.

- Define command authority, communications protocol, test boundary, exclusion zones and emergency stop arrangements.
- Use a test configuration certificate showing vehicle identity, software version, equipment fitted, temporary modifications and known defects.
- Ensure temporary instrumentation does not compromise insulation, fire protection, structural integrity, braking, EMC or passenger safety.
- After testing, remove temporary equipment, reinstate covers and repeat the checks affected by removal.

17. Dynamic testing, commissioning and acceptance

17.1 Dynamic test programme

Dynamic tests should be performed at an appropriately authorised test facility or on a route under the required permissions. The programme is designed by competent vehicle dynamics, braking, electrical and safety engineers. It must be aligned with the applicable standards and infrastructure conditions.

Test area	What to demonstrate	Evidence to retain
Running characteristics	Stable running, acceptable wheel/rail forces, curving, steering behaviour and response over track conditions	Test configuration, track data, wheel profile, speed, loading, sensor calibration and analysis report.
Braking	Stopping performance, brake blending, emergency brake, thermal behaviour, wheel-slide protection and degraded modes	Speed/load/gradient/adhesion conditions, measured data, calculation correlation and pass/fail statement.
Traction	Acceleration, power limitation, current collection, thermal stability, wheel slip and recovery	Power supply conditions, current/voltage, speed, temperature, software and fault logs.
Ride and noise	Passenger comfort and acoustic targets across seats, speeds and operating modes	Measurement locations, track quality, ambient conditions, HVAC state and calibrated equipment.
Aerodynamics	Pressure transients, crosswind response and equipment cooling as required	Vehicle configuration, weather, tunnel/route conditions, pressure and temperature data.
Energy	Measured energy use and regenerative energy over a defined duty cycle	Duty cycle, passenger load, weather, meter calibration and auxiliary state.
Reliability trial	Fleet/system reliability and maintainability under representative operation	Distance, service hours, cancellations, failures, delays, repeat defects and corrective actions.

17.2 Commissioning sequence

Commission the production train

01 Complete static release and confirm all safety-critical defects are closed.

02 Confirm test route, permissions, operating limits, staff competence, rescue arrangements and contingency plan.

03 Begin with low-speed movement and basic control/brake checks in the approved test area.

04 Progress through the test envelope in planned steps. Do not increase speed or load until the previous test point is reviewed and released.

05 Execute braking, traction, running dynamics, noise, ride, HVAC and energy tests in the approved order.

06 Inspect wheelsets, bearings, bogies, brake equipment, structural joints and underframe after defined test stages.

07 Review anomalies promptly. Stop testing where a safety limit, unexpected trend or unassessed fault is observed.

08 Complete the reliability trial and close recurring defects through root-cause analysis.

09 Compile the final test report, as-built configuration, safety evidence and authorisation file.

17.3 Acceptance and handover

Acceptance should be based on objective evidence, not an informal demonstration. The handover pack should include the as-built bill of material, vehicle configuration, type and routine test reports, quality dossier, safety and conformity evidence, maintenance instructions, parts catalogue, diagnostic guide, training material, software baseline, warranty terms, special tools and spares recommendation.

ACCEPTANCE TIP Agree the defect classification and concession rules before factory acceptance testing. Define which defects block dispatch, which may be accepted with a time-bound corrective action, and who has authority to approve each category. Safety-related defects must never be hidden in a general punch list.

18. Reliability, maintenance and 30-40 year support

18.1 RAMS from concept

Reliability, Availability, Maintainability and Safety (RAMS) should be managed across the complete lifecycle. Define the service mission, operational availability target, reliability allocation, maintenance concept and safety risk acceptance method. Use FMEA/FMECA, fault-tree analysis, hazard logs and maintainability analysis at the level appropriate to each system.

18.2 Design for maintainability

- Provide safe access to inspection points without removing unrelated equipment.
- Use replaceable line-replaceable units for electronics, fans, pumps, sensors and other modular equipment where lifecycle economics justify them.
- Standardise fasteners, connectors, filters, lamps and service consumables where this does not compromise function or safety.
- Design cable and pipe routes so technicians can trace them and replace sections without destructive removal of interiors.
- Provide lifting points, component mass labels and removal paths for heavy equipment.
- Create condition-based maintenance rules that retain mandatory safety inspections and statutory checks.

18.3 Maintenance data and digital twin

The vehicle digital record should link each serialised component to supplier, batch, install date, test evidence, maintenance action, software compatibility and removal history. A digital twin need not be a photorealistic 3D model; its value is the controlled connection between the as-built asset, engineering configuration, operating data and maintenance records.

18.4 Obsolescence and mid-life refresh

Build obsolescence control into the programme

01 Identify components with short commercial lifetimes: processors, displays, storage, batteries, power modules, network devices and sensors.

02 Require suppliers to provide lifecycle status, change notification and last-time-buy policy.

03 Prefer open, documented interfaces and replaceable modules over bespoke electronics where possible.

04 Maintain a software bill of materials and cybersecurity support policy for digital components.

05 Plan a mid-life refresh around modular replacements, interior refurbishment, energy improvements and diagnostic upgrades.

06 Preserve the original vehicle configuration and change history so later upgrades can be assessed safely.

19. Late-2020s design tips: performance, appearance and sustainability

19.1 Exterior styling that follows engineering

A late-2020s visual language can be contemporary without becoming disposable. Use a confident front-end, continuous glazing where practical, sculpted but manufacturable body transitions, flush lighting, carefully integrated sensors and clean side surfaces. Avoid adding fragile decorative parts that multiply fasteners, water traps and repair operations.

- Use a single datum-driven side profile to control window, beltline, door and panel alignment across the complete fleet.
- Keep headlight, marker light, camera, sensor and demister locations within an agreed optical and maintenance envelope.
- Use replaceable nose and corner modules where crash design and repair strategy support them.
- Design access panels to align with real maintenance tasks, not arbitrary styling lines.
- Plan livery and brand graphics in the digital mock-up to avoid poor fit at door seams and service hatches.

19.2 Interior design language

- Use a calm material palette with limited, durable finishes; reserve high-wear surfaces for replaceable components.
- Integrate lighting into ceiling and seat geometry while keeping drivers and emergency systems accessible for replacement.
- Use clear zoning: quiet areas, work tables, family areas, luggage zones and accessible spaces, if required by the service concept.
- Use acoustic treatment at the source: resilient mounts, balanced rotating equipment, airflow management and well-sealed panels.
- Design cleaning access into seat bases, table mountings, floor edges and toilet enclosures.

19.3 Sustainability and circular manufacture

Measure environmental impact at vehicle and fleet level. Lightweighting can reduce operational energy but may increase material processing complexity. Compare embodied carbon, operational energy, maintenance materials, refurbishment and end-of-life recovery. Prefer clear material identification, reversible joints where suitable, repairable modules and recyclable material streams.

19.4 Ten avoidable design mistakes

Mistake	Better practice
Styling is frozen before the structure	Freeze the engineering envelope first; develop styling concurrently under configuration control.
Too many bespoke components	Standardise proven components and keep bespoke parts where they deliver clear lifecycle benefit.

Mistake	Better practice
No access to hidden equipment	Perform service/removal-path reviews in the 3D mock-up before release.
Unmanaged mass growth	Run a live mass and axle-load budget with formal change approval.
Different suppliers use different datums	Publish master vehicle datums and interface-control documents.
Testing begins after production is complete	Build the verification matrix and test rigs at concept stage.
Inspection records are detached from serial numbers	Capture evidence at point of manufacture in the digital traveller.
Fire compliance treated as a material certificate only	Qualify the actual finished assembly and installed configuration.
Diagnostic data is collected without fault logic	Define actionable fault codes, thresholds, owner and maintenance response.
No repair philosophy for new materials	Design repair, inspection and depot capability before selecting a material or joining method.

20. Factory implementation roadmap

A new rolling-stock manufacturer should build capability in phases. The sequence below avoids investing in a large production facility before product definition, supplier strategy and certification route are credible.

Phase	Main work	Exit criteria
0. Programme definition	Choose vehicle market, customer, route, operating need, design basis, industrial strategy and business case.	Approved DBM, route interface pack, target product architecture and initial risk register.
1. Engineering capability	Recruit chief engineer, structures, bogie, electrical, brake, RAMS, software, manufacturing and quality leads. Select design tools and PLM.	Signed engineering management plan, design review plan and requirements database.
2. Supply chain qualification	Identify body material, wheelset, bogie, traction, brake, door, HVAC, glazing and interior suppliers. Audit critical sources.	Approved supplier list, technical specifications and supplier quality agreements.
3. Pilot production cells	Build representative carbody/bogie/electrical pilot cells. Qualify joining, fixtures, metrology and test processes.	First-article acceptance, qualified special processes and stable work instructions.
4. Prototype vehicle	Manufacture a complete prototype using production-intent materials, tooling and components.	Integrated static test release and type-test-ready configuration.
5. Type test and authorisation	Run structural, dynamic, brake, electrical, EMC, fire, software and route compatibility evidence programme.	Completed conformity/safety evidence and required authorisation.
6. Rate production	Complete production readiness review and ramp output using first-pass quality, supplier capacity and stable configuration.	Repeatable line performance and approved vehicle acceptance dossiers.
7. Fleet support	Operate reliability growth, warranty, maintenance analytics, spares and obsolescence controls.	Closed-loop service feedback and funded lifecycle support plan.

20.1 Suggested manufacturing organisation

Function	Accountability
Chief Engineer / Design Authority	Owns design integrity, technical decisions, design baseline and engineering approvals.
Vehicle Systems Engineering	Owns interfaces, requirements, integration and technical architecture.
Structures and Materials	Owns body, bogie structure, materials, fatigue, joining and repair engineering.
Electrical and Software	Owns power architecture, controls, TCMS, software assurance, EMC and cybersecurity.
Manufacturing Engineering	Owns process design, fixtures, routings, industrialisation, production capability and work instructions.
Quality and Supplier Quality	Owns QMS, ITP, audits, inspection independence, non-conformance and records.
RAMS and Safety	Owns hazard management, reliability allocations, safety evidence and lifecycle risk.
Test and Commissioning	Owns verification plan, test rigs, vehicle test, dynamic test coordination and reports.
Aftermarket / Fleet Support	Owns maintenance documentation, spares, warranty, training, obsolescence and service feedback.

Appendix A. Inspection and test plan template

Copy this table into the project ITP and populate it for each part or operation. The acceptance reference must always point to the controlled drawing, specification, procedure or applicable standard revision.

Step	Characteristic / operation	Method and equipment	Acceptance reference	Extent / frequency	Point / record
A-010	Incoming material identity	Certificate review + visual / barcode	Material specification, PO, drawing rev.	Each batch / heat	Hold; incoming inspection report
A-020	Cut profile geometry	Calibrated profile gauge / CMM	Released drawing tolerance	First-off + defined sample	Witness; dimensional report
A-030	Structural joint preparation	Visual + gap / fit-up gauge	Released WPS and joint detail	Each joint before joining	Hold; fit-up checklist
A-040	Production weld	Process monitoring + weld map	Qualified WPS / EN 15085 requirements	100% process record	Surveillance; weld log
A-050	Weld inspection	VT and specified NDT	Approved NDT plan / acceptance criteria	As specified by weld class	Hold; NDT report
A-060	Carbody geometry	Laser tracker / CMM	Released datum and tolerance plan	Defined points on each shell	Hold; survey report
A-070	Electrical installation	Inspection + electrical tests	Electrical test specification	Each vehicle / circuit	Hold; electrical test report

Step	Characteristic / operation	Method and equipment	Acceptance reference	Extent / frequency	Point / record
A-080	Final function	Approved test procedure	System requirement / test limits	Each vehicle	Hold; test certificate

ITP governance: identify the notice period for witness points, the names/roles permitted to release hold points, the handling of missed witness attendance, and the exact NCR route for failed results. Do not leave "as per standard" as the only acceptance criterion where the standard requires project-specific values.

Appendix B. Manufacturing traveller templates

B1. Carbody traveller

Field	Required entry
Vehicle / car number	[Unique vehicle and car identifier]
Shell serial number	[Unique structural shell ID]
Drawing baseline	[PLM release ID and revision]
Material traceability	[Heat / batch / profile IDs and certificate references]
Fixture ID / status	[Fixture number, verification date, calibration status]
Cutting / machining	[Program revision, machine ID, first-off report]
Joining records	[Weld IDs, WPS, operator IDs, date/time, parameter record]
NDT	[Method, extent, inspector qualification, report IDs]
Dimensional survey	[Instrument ID, calibration, datum setup, deviations]
Rework / concessions	[NCR and authorised disposition references]
Release	[Quality sign-off, engineering sign-off, date]

B2. Bogie traveller

Field	Required entry
Bogie serial / orientation	[Unique ID and vehicle location]
Frame material	[Steel grade, heat IDs, certificates]
Frame welds	[Weld map, WPS, welder, inspection reports]
Frame geometry	[Datum report, critical mount dimensions]
Wheelsets	[Wheel and axle serials, product certs, assembly reports]
Bearings / suspension	[Part IDs, batch/serial, certificates, installation checks]
Brake / traction equipment	[Component IDs, installation and functional records]
Fastener control	[Joint ID, method, tool ID, result]
NDT / final inspection	[Reports, inspector, closure status]

Field	Required entry
Bogie release	[Quality and engineering approvals]

B3. Electrical installation traveller

Field	Required entry
Vehicle / system	[Vehicle, car, system and drawing revision]
Equipment identity	[Supplier, part number, serial number, software version]
Cable / harness IDs	[Cable schedule references and continuity result]
Routing inspection	[Support, bend radius, segregation, fire barrier, chafe protection]
Crimps / terminations	[Tool/applicator ID, first-off verification, operator]
Bonding / insulation	[Test procedure, instrument ID, conditions, measured result]
Functional test	[Procedure ID, software baseline, result, fault log]
As-built update	[Drawing redlines / PLM update reference]
Release	[Electrical engineer and quality sign-off]

Appendix C. Final vehicle acceptance checklist

- ☐ Design baseline, configuration index and approved deviations are complete.
- ☐ All required material and component certificates are present and traceable.
- ☐ Carbody structural calculations, weld records, NDT and dimensional survey are accepted.
- ☐ Bogie frames, wheelsets, bearings and suspension have complete release dossiers.
- ☐ All safety-critical fasteners and joints have completed the approved verification method.
- ☐ High-voltage and low-voltage installations have passed approved electrical tests.
- ☐ Protective bonding, earthing, insulation and electrical protection have been verified.
- ☐ Brake calculations, static tests and dynamic brake tests are accepted.
- ☐ Door endurance, obstruction detection, emergency release and isolation tests are accepted.
- ☐ Fire/smoke evidence exists for the actual installed material and component configurations.
- ☐ HVAC, lighting, passenger information, toilet and accessibility systems are accepted.
- ☐ TCMS and all onboard software versions/checksums are recorded and configuration-controlled.
- ☐ EMC, shock/vibration and environmental evidence is complete where applicable.
- ☐ Running characteristics and route compatibility evidence is accepted.
- ☐ All safety-critical defects are closed; remaining defects have approved classification and disposition.
- ☐ Maintenance manuals, parts catalogue, training, special tools and spares plan are delivered.
- ☐ As-built drawings, wiring diagrams, software baseline and asset register are complete.
- ☐ Final manufacturing quality dossier and acceptance certificate are signed.

Appendix D. Reference standards and compliance links

The standards below are a starting register for a GB passenger rolling-stock programme, checked against public publisher or government catalogue information on 2 November 2026. Applicability depends on vehicle category,

contract, route, design, project dates and national rules. Obtain the full licensed standard text, review every normative reference and amendment, and create a project-specific compliance matrix before design approval.

01. GB authorisation / interoperability: NTSN: Rolling Stock - Locomotive and Passenger (LOCPAS), Issue 2, 2 May 2025 - GB requirements for the locomotive and passenger rolling-stock subsystem. Check transitional provisions and related NTSNs/NTRs.

02. Authorisation: ORR: Authorising vehicles and infrastructure / interoperability - ORR explains authorisation before new, upgraded or renewed rolling stock is placed into use. Interoperability compliance does not replace wider safety duties.

03. Railway QMS: ISO 22163:2023 - Railway quality management system requirements, based on ISO 9001 with railway-specific additions.

04. Carbody structure: BS EN 12663-1:2010+A2:2023 - Structural requirements for locomotive and passenger rolling-stock bodies.

05. Crashworthiness: BS EN 15227:2020+A1:2024 - Crashworthiness requirements for rail vehicles. Confirm applicable vehicle category and scenarios.

06. Bogie frames: BS EN 13749:2021+A1:2023 - Method of specifying structural requirements of bogie frames.

07. Running characteristics: BS EN 14363:2016+A2:2022 - Testing and simulation for acceptance of railway vehicle running characteristics.

08. Railway welding: BS EN 15085-2:2020+A2:2025 - Requirements for the welding manufacturer; check the full EN 15085 series and project weld classification.

09. Fire behaviour: BS EN 45545-2:2020+A1:2023 - Fire behaviour requirements for materials and components on railway vehicles.

10. Entrance systems: BS EN 14752:2025 - Passenger bodyside entrance systems and associated tests.

11. RAMS: BS EN 50126-1:2017+A1:2024 - Generic RAMS lifecycle and process. Consider related parts and safety methods.

12. Software: BS EN 50716:2023 - Requirements for software development for railway programmable electronic systems, including onboard rolling stock.

13. Onboard electronics: BS EN 50155:2026 - Railway rolling-stock electronic equipment. Verify the final adopted text and project transition.

14. Shock and vibration: BS EN IEC 61373:2026 - Shock and vibration tests for rolling-stock equipment.

15. EMC apparatus: BS EN 50121-3-2:2016+A1:2019 - Electromagnetic compatibility requirements for rolling-stock apparatus; review all related EN 50121 parts.

16. Wheelset: BS EN 13260:2025 - Wheelset product requirements.

17. Axles: BS EN 13261:2024 - Axle product requirements.

18. Wheels: BS EN 13262:2026 - Wheel product requirements.

19. Brake calculations: EN 14531-1:2015+A1:2018 - General algorithms for brake performance calculations. It does not itself prescribe the vehicle performance requirement.

Compliance workflow

Create the project compliance matrix

01 Identify all applicable legal requirements, NTSNs, NTRs, standards, customer requirements and infrastructure manager rules.

02 For each requirement, classify it as applicable, not applicable with rationale, open point, specific case, deviation or project-specific requirement.

03 Allocate each applicable clause to an owner and a verification evidence item.

04 Identify independent assessment / notified or designated body involvement required for the chosen authorisation route.

05 Assess changes and deviations for safety, interoperability and authorisation impact.

06 Close the matrix with objective evidence before submitting the authorisation application and before fleet handover.

Appendix E. Glossary and engineering abbreviations

Term	Meaning
CAE	Computer-Aided Engineering
CAD	Computer-Aided Design
CMM	Coordinate Measuring Machine
CMP	Carbody Manufacturing Plan
DBM	Design Basis Memorandum
ECR / ECO	Engineering Change Request / Engineering Change Order
EMC	Electromagnetic Compatibility
EMU	Electric Multiple Unit
FMEA / FMECA	Failure Mode and Effects Analysis / Criticality Analysis
FEA	Finite Element Analysis
FSW	Friction Stir Welding
HVAC	Heating, Ventilation and Air Conditioning
ICD	Interface Control Document
ITP	Inspection and Test Plan
MES	Manufacturing Execution System
NCR	Non-Conformance Report
NDT	Non-Destructive Testing
PLM	Product Lifecycle Management
QMS	Quality Management System
RAMS	Reliability, Availability, Maintainability and Safety
TCMS	Train Control and Management System
UT / MT / PT / RT	Ultrasonic / Magnetic Particle / Penetrant / Radiographic Testing
WPS	Welding Procedure Specification

Closing note

A high-quality late-2020s train is the result of disciplined systems engineering, capable suppliers, qualified processes, precise manufacturing, evidence-led inspection and a strong lifecycle support strategy. The practical test of excellence is repeatability: the factory should be able to manufacture successive vehicles to the same controlled configuration, demonstrate conformity with objective evidence and support the fleet safely for decades.

RELEASE-TO-BUILD RULE No production drawing, weld procedure, press-fit parameter, torque value, brake acceptance limit, electrical test voltage, structural allowable or dimensional tolerance is valid solely because it appears in this handbook. Production values must be released by the responsible design authority in the controlled project documentation.